

NET PRIMARY PRODUCTION AND CARBON ALLOCATION PATTERNS OF BOREAL FOREST ECOSYSTEMS

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Abstract. The three objectives of this paper were: to summarize net primary production (NPP) and carbon allocation patterns for boreal forests, to examine relationships between climatic and biological variables and NPP, and to examine carbon allocation coefficients for all boreal forests or types of boreal forests that can be used to estimate NPP from easily measured components of NPP. Twenty-four Class I stands (complete NPP budgets) and 45 Class II boreal forest stands (aboveground NPP [NPP_A] and budget only) were identified. The geographic distribution of the Class I stands was not uniform; 46% of the stands were from two studies in North America, and only one stand was from the important larch forests of Eurasia. Total (above- and belowground) net primary production (NPP_T) ranged from 52 to 868 g C·m⁻²·yr⁻¹ and averaged 424 g C·m⁻²·yr⁻¹. NPP_A was consistently larger for deciduous than for evergreen boreal forests in each of the major boreal regions, especially for boreal forests in Alaska. Belowground net primary production:total net primary production ($NPP_B : NPP_T$) ratios were consistently larger for evergreen (0.36) than deciduous (0.19) boreal forests. NPP of different-aged stands in age sequence varied from 44% to 77%, a magnitude equal to or greater than that of climatic factors or vegetation type. NPP and NPP_A were positively correlated ($r^2 = 0.66$ – 0.68) to mean annual aboveground increment for Class I stands, and this empirical relationship explained 81% of the observed variation of NPP_A for Class II stands. These robust relationships provide an approach for increasing the number and spatial coverage of boreal forest NPP data needed to evaluate NPP estimates from ecosystem models. Notable deficiencies of boreal forest NPP data were ground layer vegetation and belowground NPP data, NPP data for boreal forest age sequences, and NPP data for boreal larch ecosystems in Eurasia.

Key words: aboveground net primary production; belowground net primary production; boreal forests; carbon allocation; litter production; net primary production.

INTRODUCTION

Boreal forests are believed to play an important role in the global carbon budget for several reasons. Boreal forests and woodlands cover ~14.5% of the land surface and contain a disproportionately large amount of carbon in the soil compared to other biomes (Melillo et al. 1993). Boreal vegetation and soil together contain ~300 Pg of carbon—equivalent to ~50% of the carbon in the atmosphere. The large amount of carbon stored in the cold or frozen soils makes boreal forests extremely sensitive to climate change.

Recent studies suggest that high-latitude temperate and boreal ecosystems are currently a net carbon sink (Tans et al. 1990, Ciais et al. 1995, Keeling et al. 1996). However, climate change models suggest that boreal forests

will experience the greatest warming of any forest biome, with the greatest increases occurring in the continental interiors (IPCC 1996). Goulden et al. (1998) concluded that the warmer-than-normal summer temperatures in northern Manitoba were responsible for a mature black spruce (*Picea mariana*) ecosystem switching from a carbon sink to a carbon source. The large size of the boreal forest, the large amount of carbon contained in the soil, the sensitivity of net primary productivity (NPP) and net ecosystem exchange (NEE) to small climatic variation, and anticipated climate warming make the boreal forest biome a key biome to understand and represent correctly in global carbon models.

Net ecosystem exchange, the net exchange of CO₂ between terrestrial ecosystems and the atmosphere, is the balance between two major processes: net primary production and heterotrophic respiration. Net ecosystem exchange, as well as energy and water vapor flux, can be measured using the micrometeorological technique known as eddy covariance, but the cost of the system and the need to closely monitor it currently limit its widespread use. Therefore, quantifying the role of

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TABLE 1. Summary of site characteristics and net primary productivity (NPP) methodology for the Class I boreal forest sites.

Country	Site†	Latitude, longitude	Mean annual precipitation (mm)	Mean annual temperature (°C)
Saskatchewan, Canada	BOREAS SSA	53.99 N, 104.99 W	405	-1.1
Manitoba, Canada	BOREAS NSA	55.99 N, 98.99 W	536	-4.6
China	Daxing'anling	50.83 N, 121.50 E	500	-5.4
Finland	Ilomants	62.85 N, 30.88 E	na	na
Russia	Karelia	62.00 N, 34.00 E	650	2.2
Russia	Tomsk	58.00 N, 83.00 E	501	na
Russia (Siberia)	Irkutsk	53.00 N, 103.00 E	na	na
Sweden	Jädraås	60.82 N, 16.50 E	731	3
United States	Alaska	64.75 N, 148.25 W	269	-3.5

Note: The abbreviation "na" means "data not available."

† See *Methods: Description of Class I study sites* for an explanation of site abbreviations.

‡ Method abbreviations are as follows: A = allometry; CP = clip plots; MP = mesh plots; M = minirhizotrons; LC = longevity census; L = litterfall; SC = soil cores; and F = flux.

§ Leaf area index (LAI) was estimated from foliage biomass using specific leaf area values for *Pinus*, *Picea*, and *Populus*.
|| Sources: 1, Gower et al. 1997; 2, Steele et al. 1997; 3, Han 1994; 4, Finer 1989; 5, Kazimirov et al. 1977; 6, Gabeev 1990; 7, Schulze et al. 1995; 8, Linder and Axelsson 1982; 9, Flower-Ellis and Persson 1980; 10, Ruess et al. 1996; 11, Oechel and Van Cleve 1986.

¶ Fine root NPP was estimated assuming fine root turnover of 1 yr based on earlier research (Kalela 1949, Heikurainen 1955).

boreal forests in the global carbon cycle will require several coordinated approaches involving ecophysiological, micrometeorological, and remote sensing measurements and ecosystem modeling (Canadell et al. 2000, Running et al. 1999). Eddy flux towers provide continuous flux measurements of energy, water, and carbon that can be compared to model estimates to determine if the ecosystem process model is correctly simulating the effect of environmental and ecophysiological factors on water, energy, and carbon flux at various time scales (diel, seasonal, etc.). Critical field ecophysiological measurements such as vegetation cover and leaf area index (LAI) can be compared to indirect estimates derived from remotely sensed reflectance data. An ecosystem process model, driven by remotely sensed data, can be used to integrate NEE estimates across space and time; however, there is a great need to validate components of the carbon budget.

Net primary production is an important parameter in global carbon models because it can be validated with field measurements and it is an important component of net exchange of carbon dioxide between terrestrial ecosystems and the atmosphere. Comparing measured NPP estimates to ecosystem model predictions of NPP for boreal forests covering a broad range of environmental and ecological conditions is useful. Good agreement between measured and simulated NPP increases the confidence in the NEE estimates derived from ecosystem carbon budget models, while unsatisfactory agreement prompts scientists to better understand the effects of environmental and ecological controls on NPP and NEE.

Several environmental, structural, and ecological characteristics that influence NPP are unique, or very important, to boreal forests. The depth to permafrost affects the length of the growing season suitable for root growth (Steele et al. 1997) and soil processes that regulate water

and nutrient availability. Cold or frozen soils restrict decomposition, leading to deep accumulations of surface organic layers that sequester nutrients, resulting in low nutrient availability in most boreal forests (Flanagan and Van Cleve 1983).

The canopy architecture of boreal forests differs from many other forests in three respects: LAI is smaller for boreal forests than many other forests (Gower et al. 1997, 1999), the foliage distribution is highly clumped (Chen et al. 1997, Kucharick et al. 1997), and soil surface is often covered bryophytes. These characteristics influence light interception, and in turn, NPP. Nutrient availability influences LAI of forests (Landsberg and Gower 1997).

Disturbance, a natural component of the ecology of boreal forests, also influences NPP. Natural disturbances include fire (Larsen 1980, Dyrness et al. 1986, Goldammer and Furyaev 1997) and insect and pathogen infestation (Kurz et al. 1995). Human-related disturbances such as land clearing, logging, and pollution are also important (Kurz et al. 1995). Age-related changes in NPP are well documented (Sprugel 1985, Gower et al. 1996, Ryan et al. 1997), therefore, changes in disturbance regime (frequency or intensity) may affect the capacity of the boreal forests to sequester carbon.

The overall objective of this study was to compare NPP budgets for boreal forests. The availability and completeness of data differ among the major boreal forest regions. One objective was to identify boreal regions that have little or no NPP data. A second objective was to compile as many "complete" NPP budgets for boreal forests as possible and determine if there are constant carbon allocation coefficients that can be used to adjust "incomplete" NPP data for numerous stands that cover a broader range of environmental and ecological conditions. The motivation for the second objective was that the number

TABLE 1. Extended.

LAI§	Method‡					Source
	Trees	Understory	Moss	Coarse roots	Fine roots	
A	A	CP	MP	A	M	1, 2
A	A	CP	MP	A	M	1, 2
na	A	none	none	A	LC	3
A	A	none	none	A	SC¶	4
A	A	none	none	A	LC	5
A	A	none	none	A	LC	6
A	A	none	na	A	LC	7
A	A	none	none	A	SC	8, 9
na	A	none	F	none	SC	10, 11

and geographic distribution of boreal forest NPP budgets is inadequate to rigorously test model estimates of NPP (Kucharik et al. 1999). A third objective was to use regression analysis to elucidate climatic and ecological controls on NPP and carbon allocation ratios for boreal forests. Such algorithms are useful in developing carbon allocation rules for ecosystem models (Running and Gower 1991).

METHODS

Boreal forest biome

Boreal forests occupy a nearly contiguous circumpolar band in the northern hemisphere. Estimates of the total area of boreal forest vary from 12.2 to 15.9×10^8 ha, depending upon the definition used to define boreal forest (Table 3). The southern boundary occurs as far south as 50° N in continental regions and as far north as 60°–65° N in maritime regions. The largest contiguous area of boreal forest extends from Scandinavia to eastern Siberia, and the second largest boreal forest region is a 500–600 km wide zone that extends from eastern Canada to northern British Columbia, Canada and Alaska, USA (Landsberg and Gower 1997). The southern boundary of the boreal forest transitions into cool temperate conifer, deciduous, or mixed forests in eastern North America and Europe, while in other regions there is a relatively abrupt transition from boreal forest to prairie or dry steppe (Walter 1979).

The climate of boreal forests varies with latitude and geographic location. The latitudinal range of boreal forests is highly variable. Boreal forests at the southern boundary have ~120 d that the air temperature is >10°C vs. 30 d at the northern boundary of the boreal forest. The boreal maritime climate has smaller temperature amplitude than the boreal continental climate, which in extreme locations can range from >30°C in the summer to –70°C in the winter.

The floristic diversity and composition of boreal forests varies with climate and soil, but the composition of tree species is simple compared to other forest biomes (Landsberg and Gower 1997). There are only 9 dominant tree species in North America (Payette 1992) and 14 in Fennoscandia and the former Soviet Union (Nikolov and Helmsaari 1992). With the notable exception of the larch

forests in Eurasia, the Class I and II data used in our paper include most of the dominant boreal forest ecosystems in each of the major boreal regions.

Boreal larch forests occur in North America and Europe, but they do not attain the importance of larch forest in Eurasia (Gower and Richards 1990). Siberian larch (*Larix siberica* Ledeb) in central Siberia and *L. gmelinii* (Rupr.) in far eastern Russia and northeast China are the dominant tree species on >50% of all forest lands in north-eastern Asia (Krankina and Ethington 1995, Kukuev et al. 1997).

Potential errors and bias of NPP measurements

In theory NPP can be estimated as the difference between gross primary production (GPP) and autotrophic respiration (R_A):

$$\text{NPP} = \text{GPP} - R_A \quad (1)$$

However, NPP is not estimated using this approach because GPP cannot be measured directly and R_A is difficult to measure, especially in large-statured or multi-species forests. The studies reported in this paper measured NPP using the following equation:

$$\text{NPP} = \sum B_i + H \quad (2)$$

where B_i is the annual biomass production of the i th tissue (e.g., stem, branch, foliage, reproductive tissue, coarse and fine roots) and H is herbivory. Allometric equations are used to estimate the annual biomass of each component (except for fine roots), and biomass increment is calculated as the difference between measurement periods. A detailed description of this approach and potential pitfalls of using generic vs. site-specific allometric equations is provided by Gower et al. (1999). In the original papers NPP was expressed in either units of dry biomass or carbon per unit area and unit time. We used carbon grams per square meter per year and converted all biomass values to carbon using ratios of 0.5 for all woody tissue and 0.45 for foliage and fine roots.

There are several potential sources of error associated with NPP measurements. Some scientists estimated foliage production from leaf litterfall biomass data because new foliage biomass allometric equations were not available. This approach assumes the foliage mass of the can-

TABLE 2. Summary of stand characteristics and biomass distribution for the Class I boreal forest sites.

Country and site†	Dominant species	Stand age (yr)	Trees (no./ha)	Basal area (m ² /ha)
Saskatchewan, Canada BOREAS SSA	<i>Picea mariana</i>	115	5900	27
	<i>Pinus banksiana</i>	65	1190	17
	<i>Populus tremuloides</i>	67	1280	34
Manitoba, Canada BOREAS NSA	<i>Picea mariana</i>	155	5450	36
	<i>Pinus banksiana</i>	63	1310	13
	<i>Populus tremuloides</i>	53	1960	27
China Daxing'anling	<i>Larix gmelinii</i>	40	na	12
Finland Ilomants	<i>Picea abies</i>	100	731	28
	<i>Pinus sylvestris</i>	45	1108	20
	<i>Pinus sylvestris</i>	50	357	7
	<i>Betula pubescens</i>	50	579	12
Russia Karelia	<i>Picea abies</i>	63	3700	36
Tomsk	<i>Pinus sylvestris</i>	59	1587	27
Russia (Siberia) Irkutsk	<i>Pinus sylvestris</i>	75	7643	25
	<i>Pinus sylvestris</i>	95	1707	40
	<i>Pinus sylvestris</i>	70	1698	34
Sweden Jädraås	<i>Pinus sylvestris</i>	20	na	na
	<i>Pinus sylvestris</i>	120	393	16
United States Alaska	<i>Picea glauca</i>	250	na	34
	<i>Picea glauca</i>	130	na	35
	<i>Picea mariana</i>	200	na	15
	<i>Populus/Alnus</i>	30	na	23
	<i>Betula papyrifera</i>	77	na	30
	<i>Populus balsamifera</i>	90	na	27

Notes: Abbreviations are as follows: AB = aboveground biomass; BB = belowground biomass; and MAI = mean annual increment, defined as biomass divided by stand age. Sources are summarized in Table 1. The abbreviation "na" means "data not available."

† See *Methods: Description of Class I study sites* for an explanation of site abbreviations.

‡ MAI measured in (g C·m⁻²·yr⁻¹).

opy is in steady state (i.e., new foliage production $\approx$ leaf litterfall). Using leaf litterfall to estimate foliage production for an aggregating evergreen forest may underestimate new foliage production because new foliage production exceeds leaf litterfall during early stand development. Assuming "steady state" may be erroneous, especially for forests subject to drought and windstorms (Linder et al. 1987, Grier 1988).

The most common bias of the NPP data we encountered was the exclusion of one or more vegetation components, the most common being fine root and mycorrhizae (Vogt et al. 1996). We included only direct estimates of belowground net primary production (NPP_B) in this paper. We are unaware of any studies that estimated mycorrhizae NPP for boreal forests, therefore all the total NPP estimates are likely to be underestimates. We are uncertain of the error introduced by the omission of mycorrhizae NPP, but Vogt et al. (1982) estimated mycorrhizae NPP comprised 15% of NPP in a cold temperate conifer forest.

Few studies quantified the loss of NPP from herbivory in forests. The few studies that have quantified NPP have concluded that <10% of NPP of forests is consumed, except during insect outbreaks (Schowalter et al. 1986). The treatment of tree mortality in NPP es-

timates is also a problem (Binkley and Arthur 1993, Gower et al. 1999). Few of the studies from which we compiled NPP data described how they measured mortality.

Data overview and analysis

The main criterion for including published NPP data were adequate documentation of the measurements and completeness of NPP budget (above- and belowground NPP for trees, understory, and ground cover vegetation). Each stand was classified into one of three classes, depending upon completeness of NPP data and ancillary site and stand characteristics. Class I stands contain above- and belowground biomass and NPP data for overstory, understory, and in most cases, ground cover if it was deemed to be important. We excluded studies that calculated NPP_B as a fraction of aboveground biomass or aboveground net primary productivity (NPP_A), although there were several studies where it was unclear what methodology was used to estimate NPP_B. Class II stands lack NPP_B data, and in some studies understory NPP is missing. Class III stands lacked complete NPP_A budgets or had very sketchy information on NPP methodology and/or site

TABLE 2. Extended.

AB (g C/m ²)	BB (g C/m ²)	MAI‡	
		Above-ground	Total
4924	1104	43	52
3455	780	53	65
9334	1356	139	160
5721	1340	38	47
2899	733	46	58
5695	1299	107	132
2644	540	66	80
5032	2305	50	74
1649	727	37	53
2078	678	42	55
1356	408	27	35
3844	1127	61	79
8081	1793	136	174
5553	2910	74	113
8656	3215	91	125
6955	1980	99	128
572	220	29	40
2992	1004	25	33
7193	1392	48	57
7172	829	55	62
1526	599	10	14
6212	1070	207	243
7997	1042	104	117
6497	1443	72	88

characteristics. A brief description of the Class I study sites is provided below. Site characteristics and NPP measurement approaches used for each site are summarized in Table 1. Stand characteristics, carbon distribution in above- and belowground vegetation, and mean annual biomass increment are summarized in Table 2. A select number of site characteristics and NPP_A data for Class II stands are summarized in Appendix B. We attempted to conduct a thorough review of the literature and hope any omissions are minimal.

Description of Class I study sites

Canada: BOREal Ecosystem Atmosphere Study (BOREAS).—NPP data for mature jack pine (*Pinus banksiana* Lamb.), trembling aspen (*Populus tremuloides* Michx.), and black spruce (*Picea mariana* [Mill.] BSP) continental boreal forests in Canada were obtained from the two BOREAS study areas (Sellers et al. 1997). The study areas are near Thompson, Manitoba and Candle Lake, Saskatchewan Canada, referred to as the northern (NSA) and southern study areas (SSA), respectively. The SSA is near the southern boundary of the boreal forest in Saskatchewan, Canada. Winters are less harsh in the SSA than in the NSA; mean January air temperatures are -19.8°C and -25.0°C for the SSA and NSA, respectively. Summers are slightly hotter in the SSA than in the NSA: mean July air temperatures are 17.6°C and 15.7°C , respectively (Gower et al. 1997). Annual precipitation averages 405

mm in Prince Albert, Saskatchewan and 536 mm at Thompson, Manitoba. Permafrost does not occur in the upper 2 m of soil in any of the stands in the SSA, but is present at 45–75 cm in poorly drained forests and fens in the NSA.

At each study area, four fixed area plots were randomly located immediately outside the footprint of the eddy flux tower in young and old jack pine stands, mature trembling aspen, and mature black spruce forests. Plot size ranged from 7.5×7.5 m for densely stocked young jack pine stands to 30×30 m for the mature aspen stands. Site-specific allometric equations were used to estimate above-ground biomass components (stem, new and old branch, and new and old foliage), sapwood volume, and stem and leaf area (Gower et al. 1997). NPP_A is calculated as the sum of annual wood (stem + branch) and foliage production for the overstory and understory. NPP of the ground cover vegetation was estimated using plastic mesh ingrowth screens. NPP_B is calculated as the sum of annual coarse root (diameter > 10 mm) + fine root production for the mature forests only. Coarse root biomass and annual increment were calculated using non-site-specific allometric equations and fine root net primary production was estimated using minirhizotrons (Steele et al. 1997).

Russia, Siberia.—We used data from 14 plots located in two study areas: Tomsk Region, Western Siberia (Gabeev 1990) and Angara River basin, in the Irkutsk Region, near Lake Baikal (Buzykin 1978). For the remainder of the paper, the two study areas are referred to as Tomsk and Irkutsk, respectively. The geographic coordinates provided in Table 1 are approximate. The data were published in the Russian literature; the manuscripts provided good descriptions of the methods used but the description of climate were vague. The climate of both sites is continental and is representative for the southern boreal region. Scots pine (*Pinus sylvestris* L.) is the dominant tree species at both sites and is an important boreal species in Siberia; *P. sylvestris* is the dominant tree on $\sim 26\%$ of all the Siberian boreal forests (Anonymous 1990). The Scots pine stands are high density and productivity.

NPP was estimated using methods adopted by the Russian participants of The International Biological Programme (IBP). The three plots at Irkutsk range in size from 0.3 to 0.4 ha. Tree stem biomass was determined from volume and density measurements. Components of the crown (small and large branches, foliage of current year and older foliage, and cones) for each species were measured on three to six medium-size trees that were destructively sampled. The mass of groundcover/understory was measured on 20 0.25-m^2 subplots in each plot. Coarse root mass was determined by excavating root systems of sampled trees, and fine root mass was estimated from soil monoliths. Roots were separated into five diameter classes ranging from >5 mm to <0.5 mm. Wood increment was calculated from measurement of annual rings at different heights along the stem. The production of branches was calculated as a sum of current year shoots and increment of older branches. The production of roots



FIG. 1. Map illustrating the location of Class I, II, and III boreal forest stands. For Class I and II sites, the values in boxes indicate multiple net primary production measurements.

was calculated for each size class from species-specific turnover rates (Orlov 1967). Leaf litterfall was measured in 10 litter traps (0.5–1.0 m²), and branch litterfall was estimated from two 4-m² plots. Litterfall data reported in this paper are the average for 3–7 yr. Tree mortality was determined by plot remeasurements 9 yr after the plots were set up.

The methods used to estimate biomass and NPP at Tomsk plots were similar to those described for Irkutsk, except root production was estimated as a constant proportion of fine root mass plus the roots removed by tree mortality based on unavailable references. The understory production was estimated as a non-site-specific fraction of understory species biomass (Gabeev 1990). The Tomsk plots qualify only marginally for a Class I site; therefore, we used the single average value for the Tomsk plots for our Class I dataset.

Russia, Karelia.—The site is located at 62.00° N, 34.00° E in the central region of the Russian boreal forest. The climate is continental. The experimental design was a 14-stand Scots pine age sequence that spanned from 22 to 185 yr. The forest in all the plots was of the same site type. The soils were well drained, and the dominant understory vegetation was *Vaccinium vitis-idaea*. Biomass and NPP measurement procedures were similar to those used at Tomsk. The equation for calculating NPP_A was tested against field data collected from a chronosequence of medium productivity pine stands in Karelia (Northwestern Russia) (Kazimirov et al. 1977). The method used to estimate root production is questionable; therefore, we used NPP_A in the analysis.

Finland, Ilomantsi.—The study was established in 1979 on three transforming mires located at Ahvensalo in Ilomantsi, eastern Finland (62.85° N, 30.88° E; 155 m above sea level). No climate data for the site were provided. Biomass, biomass increment, and nutrient cycling were measured for control and fertilized stands (Finer 1989), but only the data for control stands were included. The experimental area was first drained in the 1930s, and

supplementary drainage occurred in the 1960s and 1979. Three forest types were studied: a 40–50-yr-old Scots pine-sedge stand, a 40–60-yr-old mixed birch (*Betula pubescens*) and pine stand with some Norway spruce, and a 100-yr-old Norway spruce (*Picea abies*) stand with some birch and pine and a *Vaccinium myrtillus* understory. The average thickness of the peat layer was >1 m in the first two stands and <1 m in the third stand.

Site-specific allometric equations were developed to estimate stem, branches and leaves, stump and coarse roots, and fine root biomass. NPP_A was estimated as the sum of annual increments of stemwood + bark, branches, and foliage. Stem diameter and bark thickness were measured for all trees >5 cm at breast height (1.37 m above the soil surface) at the beginning of the 1980 growing season and at the end of the 1985 growing season. Twenty soil cores were collected to a depth of 40 cm from each stand to estimate fine root biomass. Fine root NPP was estimated from biomass measurements assuming that <1 mm thick roots are renewed annually and 1–10 mm roots renewed every 4 yr. Litterfall was collected at 2-wk intervals during the summertime for 1983 and 1984.

Sweden, Jädraås.—The NPP of Scots pine forests were determined at the Jädraås research site during the SWECON (Swedish Coniferous Forest Biome Project) which started in 1972 (Persson 1980). Above- and belowground biomass and NPP were measured for multiple years from 1973 to 1981. Replicated 20 × 20 m control, irrigation, fertilization, and irrigation + fertilization treated plots were established in the 15-yr-old stand, but complete NPP budgets were only available for the control and irrigation + fertilization treatments.

The Jädraås study site (60.82° N, 16.50° E) is located at Ivantjärnsheden, near Jädraås in central Sweden. The study area is at the far southern boundary of the boreal forest. The climate is boreal maritime and is characterized by cool summers and cold winters. Charcoal and tar burning probably dominated recent land use at Jädraås, and the site was probably completely treeless around 1850–

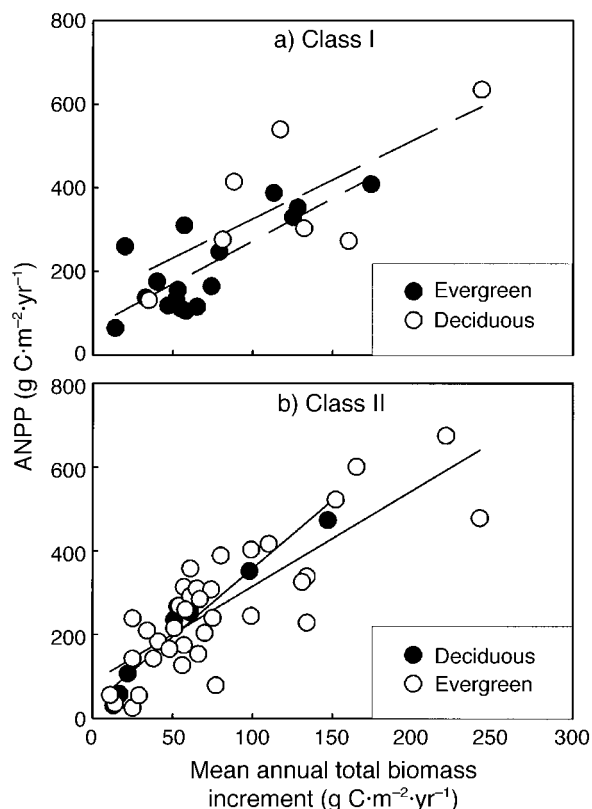


FIG. 2. (a) Relationship between mean annual total biomass increment (MAI) and aboveground net primary production (NPP_A) in Class I evergreen and deciduous boreal forest stands. The regression equation for evergreen forest is $NPP_A = 68 + 2.05(MAI)$, $r^2 = 0.62$, $P < 0.01$; the regression equation for deciduous forests was insignificant; and the equation for evergreen + deciduous forests was $NPP_A = 56 + 2.31(MAI)$, $r^2 = 0.68$, $P < 0.01$. (b) Relationship between MAI and NPP_A for Class II evergreen and deciduous boreal forest stands. The regression equation for evergreen forest is $NPP_A = 42 + 2.34(MAI)$, $r^2 = 0.81$, $P < 0.01$; for all stands, it is $NPP_A = 80 + 2.62(MAI)$, $r^2 = 0.65$, $P < 0.01$.

1860 due to charcoal consumption for smelting. The old forest stand at Jädraås was thinned around 1960.

The NPP Class I data used in this paper were from a young 20-yr-old control stand (plot "Ih II") and a 120-yr-old stand (i.e., established about 1854; plot "Ih V"). The naturally regenerated "Ih II" plot was established in 1960 with fertilization and irrigation treatments started in 1974. Data for both stands were collected in 1979.

Site-specific allometric equations were developed to estimate stem, branch, new and old foliage, and coarse root biomass from stem diameter (Flower-Ellis and Persson 1980). NPP_A was estimated as the sum of the annual production of stems, branches, and foliage. NPP_B was estimated as the sum of coarse root increment and fine root production (Persson 1978, 1980a, b) that was estimated from sequential cores and carbon budget approach (Linder and Axelsson 1982).

United States of America, Alaska.—The study area

is Bonanza Creek Experimental Forest Long Term Ecological Research site (BNZ LTER) located 20 km from Fairbanks, Alaska (64.8° N, 148° W). The climate is continental boreal. Daily temperatures average -24.5°C in January and $+17.5^{\circ}\text{C}$ in July, with an average annual temperature of -3.5°C . Average annual precipitation is 269 mm, 65% of which falls during the growing season. Potential evapotranspiration (466 mm) exceeds precipitation, suggesting water stress may limit growth (Viereck et al. 1993). Permafrost is discontinuous and is most common on poorly drained soils or sites with a north aspect.

The stands are representative of successional stages for upland sites following fire and for floodplain sites following alluvial bar formation (Ruess et al. 1996). The upland stands included 75–80-yr-old mixed stands of paper birch–trembling aspen (*Betula papyrifera* Marsh.–*Populus tremuloides* Michx.) and 105–155-yr-old white spruce (*Picea glauca* (Moench) Voss) stands. The alluvial floodplain stands included 200–40-yr-old balsam poplar–thinleaf alder (*Populus balsamifera* L.–*Alnus tenuifolia* Nutt.), 80–100-yr-old balsam poplar, 200–300-yr-old white spruce stands, and 150–250-yr-old black spruce stands. A detailed description of the sites is provided by Van Cleve et al. 1991.

Aboveground woody biomass production was calculated using published allometric equations for white spruce (Yarie and Van Cleve 1983) and allometric equations for the other species (Ruess et al. 1996). Annual woody biomass increment was calculated for 1989–1992. Total litterfall estimates were based on three 0.25-m² litter screens at each site; collections were made from 1990 to 1992. Fine root net primary production was estimated using the maximum–minimum approach. Fifteen cores were collected monthly from June through September for 1990 and 1991, except for the black spruce site, which was only sampled in 1990. Coarse root and understory NPP were not mentioned in the paper, so we assumed they were not measured. In an attempt to develop complete carbon budgets, we used published moss NPP data for black spruce, white spruce, and paper birch stands at BNZ LTER and birch–aspen stands at Chena Hot Springs Road (Oechel and Van Cleve 1986).

China, Daxing'anling.—Boreal forests in the Daxing'anling region of northeast China are part of the eastern Siberian boreal coniferous forest. Larch stands are the dominant forest ecosystems in the Daxing'anling region. The understory species composition varies with site conditions and elevation. Daily air temperature averages -20°C to -30°C in January and $+17$ – 20°C in July. The mean annual air temperature is -5.4°C . Annual precipitation ranges from 450 to 550 mm, 85–90% of which falls during the growing season, with an average humidity of 70–75%. Snow pack lasts for 5 mo and reaches 30–50 cm depth in the stands. Continuous permafrost occurs north to Genhe (50.66° N, 121.95° E), and discontinuous permafrost occurs south to Genhe. Permafrost is more common beneath *Pinus pumila*–*Larix gmelinii*, *Ledum*

TABLE 3. Approximate area of boreal forest biome.

Source	Ecosystems included from original source	Area†
Kucharik et al. (1999)	boreal evergreen	18.80
	boreal deciduous	2.29
	total‡	21.09
Wang and Polglase (1995)	no ecosystem names provided	9.00
Matthews (1983)	temperate/subpolar evergreen needleleaved	9.29
Melillo et al. (1993)	boreal woodland	6.30
	boreal forest	12.20
	total‡	18.50
Dorman and Sellers (1989)	needleleaf, evergreen	14.73
	needleleaf, deciduous	6.16
	total	20.89
Average		15.75

† Data are in millions of square kilometers.

‡ Denotes potential area.

palustre–*Larix gmelinii*, and lowland *Larix gmelinii* forests, and less developed in *Rhododendron dahurica*–*Larix gmelinii*, *Pinus sylvestris* var. *mongolica*, *Populus suaveolens*, and *Chosenia arbutifolia* forests (Xu 1998).

Allometric equations developed for *Larix gmelinii* in the Daxing'anling region were used to estimate biomass of stem, branch, foliage, and coarse root biomass. NPP_A was calculated as the sum of stem, branch, and foliage biomass increment. NPP_B was calculated as the sum of annual coarse root biomass increment and fine root net primary production.

Data analysis.—Class I stands are not a statistically representative sample of boreal forests in general or for a geographic region. In addition to potential biases, the sample size is very small. Therefore, we refrained from conducting formal statistical analyses to compare mean values of NPP and its components for regions or forest types. In general, an average and range are presented. We caution that the averages presented in the paper may be biased estimates of NPP for boreal forests.

We used regression analysis to examine the empirical relationships between components of NPP and NPP for all boreal forests and by leaf habit (e.g., evergreen vs. deciduous). We acknowledge that autocorrelation may be a potential problem when comparing a component of NPP to NPP. In an attempt to avoid this problem and test the robustness of the empirical relationships, we compared the empirical models developed for Class I stands to independent data for Class II stands. Except for mean annual biomass increment (MAI), a description of each component of NPP is provided above. Mean annual increments of aboveground and total biomass were calculated by dividing the aboveground biomass and total biomass by stand age.

All the regression analyses were done using SAS (SAS 2000). An intercept was included in the linear regression models to avoid inflating the r^2 values (Arnold and Good 1981), although we fully recognize that NPP is zero as the predictor approaches zero (i.e., zero intercept). Extrapolation beyond the range of reported values is strongly discouraged.

RESULTS

Availability and distribution of boreal forest NPP data

Few complete NPP budgets exist for boreal forests. We identified 24 Class I and 45 Class II boreal forest stands (Fig. 1). Several of the Class I stands lacked understory or coarse root NPP, but we included the stands because these two components generally comprise a small fraction of total NPP, and we would limit the analysis to 14 stands if the stands in question were excluded. The Class I data include boreal forests of contrasting climates (e.g., continental and maritime), evergreen and deciduous leaf habit, and broad-leaf and needle-leaf morphology. The inadequacy of complete NPP budgets for boreal forests is illustrated by a simple calculation. Estimates of the total area of boreal forests range from 9.00 to 21.09×10^6 km² and average 15.75×10^6 km² (Table 3). Assuming the stands are uniformly distributed throughout the boreal forest biome, which is an incorrect assumption, each stand represents $\sim 0.65 \times 10^6$ km², demonstrating the need to develop a larger NPP database of greater geographic coverage.

A second concern is that the geographic and stand age distribution of Class I boreal stands is not uniform (Fig. 1). Fifty percent of the Class I stands are from two North American studies, Alaska (Ruess et al. 1996) and central Canada (Gower et al. 1997), and an additional 17% of the stands are from one study site in Finland (Finer 1989). The age distribution of Class I–III stands is skewed towards older stands (Appendix A).

An additional 299 Class III stands were compiled, but many contain little additional data other than select components of NPP. The Class III stands encompass the entire range of boreal climates, forest types, stand ages, and other environmental and ecological variables that influence NPP. While the quality of these data may be high, the incomplete NPP budgets and lack of stand characteristics make it difficult to use the data to validate NPP estimates derived from ecosystem models.

Below we summarize the NPP budgets for boreal stands in each of the major geographic regions of the

boreal forest. The data are not intended to represent the entire region, but simply are the data that are available. In the discussion section we explore the comparison of the Class I NPP data to the Class II and III data in an effort to determine if the Class I stands are representative of the larger compilation of estimates of boreal forest NPP.

Russia

NPP_A for the five Class I evergreen conifer forests ranged from 185 to 439 g C·m⁻²·yr⁻¹ and averaged 359 g C·m⁻²·yr⁻¹ (Table 4). NPP_A for Class II evergreen conifer stands averaged 309 g C·m⁻²·yr⁻¹ and ranged from 127 to 675 g C·m⁻²·yr⁻¹, and NPP_A for Class II deciduous stands averaged 65 g C·m⁻²·yr⁻¹ (Appendix B). NPP_B values fell into two groups: extremely low values (41 to 51 g C·m⁻²·yr⁻¹) for *Picea abies* in Karelia and *Pinus sylvestris* in Tomsok and high values (279 to 495 g C·m⁻²·yr⁻¹) for *Pinus sylvestris* stands in Irkutsk. Total NPP averaged 628 g C·m⁻²·yr⁻¹ and ranged from 226 to 912 g C·m⁻²·yr⁻¹. The carbon allocation ratio of NPP_B : NPP_T averaged 0.37. We did not locate a complete NPP budget for a boreal deciduous forest, even though the deciduous conifer *Larix* spp. is an important forest ecosystem in far eastern Russia (Gower and Richards 1990).

Nordic countries

NPP_A for the six Class I evergreen conifer forests in Sweden and Finland ranged from 138 to 217 g C·m⁻²·yr⁻¹ and averaged 177 g C·m⁻²·yr⁻¹ (Table 4). NPP_A for Class II evergreen conifer stands averaged 100 g C·m⁻²·yr⁻¹ and ranged from 35 to 210 g C·m⁻²·yr⁻¹ (Appendix B). NPP_B values fell into two groups: one extremely low value (54 g C·m⁻²·yr⁻¹) and moderate values ranging from 103 to 285 g C·m⁻²·yr⁻¹. Total NPP averaged 321 g C·m⁻²·yr⁻¹ and ranged from 215 to 462 g C·m⁻²·yr⁻¹ (Table 4). NPP_B : NPP_T ratio averaged 0.45. One complete NPP budget was found for boreal deciduous forest in Scandinavia; NPP_A was similar for the deciduous and evergreen boreal forests, but NPP_B and NPP_B : NPP_T for the *Betula pubescens* stand was roughly half that of the evergreen conifer forests (Table 4).

Canada

Annual woody biomass increment, NPP_A, and NPP_B did not differ between the SSA and NSA for aspen, jack pine, and black spruce. NPP_A and NPP_T were consistently greater for deciduous and evergreen conifer stands at both the southern and northern study areas (Table 4). For example, NPP_A is 2.7 times greater for deciduous aspen stands (351 g C·m⁻²·yr⁻¹) than evergreen conifer black spruce or jack pine stands (129 g C·m⁻²·yr⁻¹), and NPP_T averaged 1.7 times more for deciduous (405 g C·m⁻²·yr⁻¹) than evergreen (233 g C·m⁻²·yr⁻¹) stands. NPP_B : NPP_T was two times larger for evergreen conifers (0.44) than for deciduous (0.13) stands.

United States

NPP ranged from a minimum of 225 g C·m⁻²·yr⁻¹ for a lowland black spruce stand to a maximum of 795 g C·m⁻²·yr⁻¹ for an upland aspen–alder stand (Table 4). NPP_A, NPP_B, and NPP_T were 2.2, 1.7, and 2.0 times larger for deciduous than evergreen conifer stands at the Bonanza Creek Long Term Ecological Research site. NPP_B : NPP_T ratio was larger for evergreen conifer (0.31) than deciduous (0.24) stands. NPP_A for Class II evergreen conifer stands averaged 94 g C·m⁻²·yr⁻¹ and ranged from 25 to 143 g C·m⁻²·yr⁻¹ (Appendix B).

China

One complete NPP budget of a *Larix gmelinii* stand was located for boreal forests in China. NPP_A, NPP_B, and NPP_T were 269, 54, and 323 g C·m⁻²·yr⁻¹, respectively (Table 4). NPP_B : NPP_T ratio was 0.17. NPP_A for Class II deciduous stands averaged 323 g C·m⁻²·yr⁻¹ and ranged from 236 to 474 g C·m⁻²·yr⁻¹ (Appendix B).

Estimating NPP_A and NPP_T indirectly

NPP_A was positively correlated to mean annual increment (MAI) of aboveground biomass ($r^2 = 0.65$, $P < 0.01$, data not shown) and mean annual total biomass increment ($r^2 = 0.68$, $P < 0.01$) for Class I stands (Fig. 2). In an effort to determine if the empirical model is robust, we compared equations derived from the Class I and Class II stands. We observed a significant positive correlation between NPP_A and mean annual aboveground biomass increment ($r^2 = 0.68$, $P < 0.01$) and the coefficients for the empirical model for Class II stands did not differ from the coefficients for the Class I stands (Fig. 2b). A significant positive relationship was also observed between NPP_A and mean annual increment of total vegetation biomass (data not shown). NPP_T was also positively correlated ($R^2 = 0.66$, $P < 0.01$) to NPP_A for the Class I stands (Fig. 3).

DISCUSSION

Relationship between environmental factors and NPP

Class I stands span a broad range of boreal environments, ranging from the mild maritime boreal climates in Sweden with a mean annual temperature of 2.4°–3°C to extreme continental boreal climates with mean annual temperatures of –4.6°C for northern Manitoba. NPP_A was not correlated to any environmental variables for Class I or II stands. However, when all stands (I–III) were included, we observed a positive correlation between NPP_A and mean annual temperature ($r^2 = 0.41$, $n = 94$, $P < 0.01$) and mean annual precipitation ($r^2 = 0.37$, $n = 114$, $P < 0.01$). There was also a significant inverse correlation between NPP_A and latitude ($r = -0.33$) for the 298 Class III stands, confirming that more northerly stands have lower NPP.

The poor correlations between the climate characteristics and NPP_A and NPP are not surprising. Environmental conditions influence NPP of boreal forests, but

TABLE 4. Summary of leaf area index (LAI, unitless) and net primary production (NPP, in carbon grams per square meter per year) for Class I boreal forest sites.

Country	Site†	Dominant species	LAI
Saskatchewan, Canada	BOREAS SSA	<i>Picea mariana</i>	6
		<i>Pinus banksiana</i>	2
		<i>Populus tremuloides</i>	3
Manitoba, Canada	BOREAS NSA	<i>Picea mariana</i>	4
		<i>Pinus banksiana</i>	2
		<i>Populus tremuloides</i>	2
China	Daxing'anling	<i>Larix gmelinii</i>	na
Finland	Ilomants	<i>Picea abies</i>	na
		<i>Pinus sylvestris</i>	na
		<i>Pinus sylvestris</i>	na
		<i>Betula pubescens</i>	na
Russia	Karelia	<i>Picea abies</i>	3
		<i>Pinus sylvestris</i>	na
Russia (Siberia)	Irkutsk	<i>Pinus sylvestris</i>	na
		<i>Pinus sylvestris</i>	na
		<i>Pinus sylvestris</i>	na
Sweden	Jädraås	<i>Pinus sylvestris</i>	1.4
		<i>Pinus sylvestris</i>	3.0
United States	Alaska	<i>Picea glauca</i>	2.4
		<i>Picea glauca</i>	2
		<i>Picea mariana</i>	2
		<i>Populus/Alnus</i>	3
		<i>Betula papyrifera</i>	4
		<i>Populus balsamifera</i>	6

Notes: Abbreviations are as follows: NPP_A = aboveground net primary production; NPP_B = belowground net primary production; NPP_T = total net primary production. Sources are summarized in Table 1. The abbreviation "na" means "data not available."

† See *Methods: Description of Class I study sites* for an explanation of site abbreviations.

characteristics such as mean annual precipitation, mean annual temperature, and latitude inadequately represent important environmental factors controlling NPP, except at the continental to biome scale. For example, NPP of the three BOREAS SSA, three BOREAS NSA, six Bonanza Creek LTER, and four Ilomants, Finland stands varied by 43%, 47%, 72%, and 39%, respectively (Table 4). Empirical studies suggest that seasonal distribution of precipitation, timing of soil thaw, nutrient availability, and cumulative growing season vapor pressure deficit influence the NPP of boreal forests (Linder and Axelsson 1982, Frolking et al. 1996, Frolking 1997). Few studies we examined provided the more detailed climate data to conduct additional analyses.

Local abiotic factors that influence NPP, such as aspect, topography, degree days, and soil type, greatly reduce the predictive power of single climate variables. Aspect influences the presence and depth of permafrost in Alaskan boreal forests. The most productive forests occur on south- and west-facing slopes because the greater radiation input increases decomposition and the rooting depth (Van Cleve et al. 1983, 1991). Also, soil type or topographic position influence NPP (Ruess et al. 1996, Gower et al. 1997). Another contributing factor is that many of the canopy structural characteristics (e.g., bud number, number of needles per fascicle, needle retention, etc.) of forests that affect canopy photosynthesis are influenced by the climate of past years.

Many of the environmental factors that influence NPP also influence the maximum leaf area index (Landsberg and Gower 1997). Therefore, LAI is an integrated measure of environmental constraints on resource availability and should be correlated to NPP (Grier and Running 1977, Gholz 1982, Fassnacht and Gower 1997). We observed a weak positive correlation between LAI and NPP_A ($r^2 = 0.20$, $P < 0.05$) for evergreen conifer boreal forests, but the relationship was insignificant for boreal deciduous forests (Fig. 4). Possible explanations for the large variation in LAI and NPP_A relationships are the poorly defined methodology and inconsistent definitions of LAI provided by the original authors. We attempted to use a consistent definition of LAI, but this was extremely difficult without better information on how LAI estimates were obtained.

Relationship between leaf habit and NPP and carbon allocation

One consistent pattern that continues to emerge as more boreal forest NPP data become available is that NPP and carbon allocation patterns differ between evergreen and deciduous forests (Gower et al. 1997). NPP_A and NPP are twofold larger for deciduous than evergreen conifer Class I stands (Fig. 5, Appendix A). Also, the fraction of NPP allocated to coarse and fine root primary production is almost twofold greater for evergreen conifers than deciduous boreal forests, corroborating initial ob-

TABLE 4. Extended.

NPP								
Tree wood	Tree foliage	Understory	Moss	NPP _A	Coarse roots	Fine roots	NPP _B	NPP _T
80	53	13	na	146	13	106	119	265
65	51	5	0	121	11	94	105	226
176	97	79	0	352	16	26	42	394
73	46	10	na	129	11	78	89	218
70	36	15	0	121	14	86	100	221
214	89	46	0	349	26	40	66	415
204	65	na	na	269	na	na	54	323
105	88	na	na	193	29	129	158	351
92	125	na	na	217	25	92	117	334
80	80	na	na	160	15	40	55	215
76	93	na	na	169	10	35	45	214
82	102	1	na	185	na	na	41	226
309	101	29	na	439	na	na	51	490
323	66	43	na	432	na	na	480	912
256	74	40	na	370	na	na	495	865
314	39	17	na	370	na	na	279	649
102	75	na	na	177	55	230	285	462
75	63	na	na	138	na	na	103	241
185	75	na	na	260	na	111	111	371
225	85	na	50	360	na	71	71	431
35	30	na	55	120	na	105	105	225
480	155	na	na	635	na	160	160	795
405	135	na	5	545	na	124	124	669
325	90	na	na	415	na	197	197	612

servations by Gower et al. (1997). The small standard errors of the average fraction of NPP allocated belowground for evergreen conifers or deciduous forests both for a geographic region and among geographic regions suggest the allocation coefficients are reasonably robust. The apparent “stable” carbon allocation coefficients imply that NPP can be approximated from NPP_A data without introducing large errors (see *Developing a boreal NPP database for ecosystem model validation*).

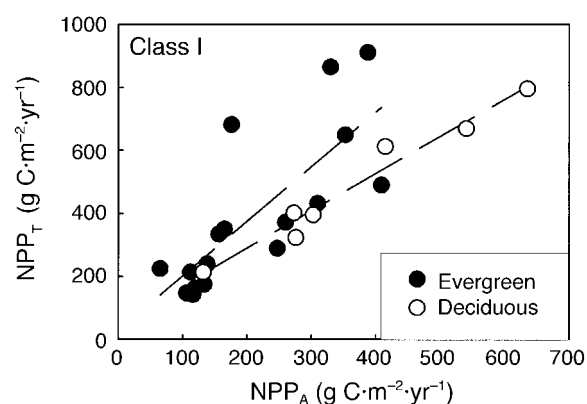


FIG. 3. Relationship between aboveground net primary production (NPP_A) and total net primary production (NPP_T) for Class I evergreen and deciduous boreal forest stands. The regression equation for deciduous stands is insignificant, the regression equation for evergreen stands is $NPP_T = 42 + 1.63(NPP_A)$, $r^2 = 0.60$, $P < 0.01$, and the regression equation for all Class I stands is $NPP_T = 114 + 1.21(NPP_A)$, $r^2 = 0.66$, $P < 0.01$.

Relationships between stand age and NPP

Age-related changes in NPP and NPP_A are a universal phenomenon (Sprugel 1985, Gower et al. 1996, Ryan et al. 1997). An important finding of this paper is that for a given geographic region, the change (i.e., decrease) in NPP during succession is of similar magnitude as the influence of environmental and soils factors on NPP. Aboveground net primary production decreased (maximum NPP_A – minimum NPP_A/maximum NPP_A) during succession by ~70% for Class II boreal *Larix gmelinii* age sequence in Siberia and a *Picea abies* age sequence (Appendix B). For two Class I boreal forest successional sequences in Alaska, NPP changed by 44% for the upland site and 77% for the alluvial floodplain site (data in Table 3 from Ruess et al. 1996).

These data clearly emphasize the need to incorporate the effects of disturbance on NPP budgets for boreal forests. Stand age distribution data have been incorporated in recent inventory-based estimates of global forest carbon storage (Schimel 1995). However, ecosystem models currently being used to simulate global terrestrial carbon budgets commonly represent the terrestrial landscape as “stable” and “mature” and do not account for the effects of disturbance on species composition, structure, and function. The assumption of steady state and omission of age-related changes in forest NPP is a concern because modeled global carbon budgets will be inaccurate unless ecosystem models correctly represent disturbance dynamics. The importance of disturbance on carbon dynamics has led to the coining of the phrase “net biome ex-

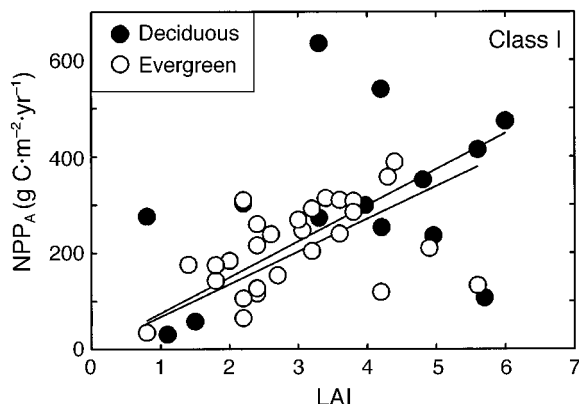


FIG. 4. Relationship between leaf area index (LAI) and aboveground net primary production (NPP_A) for Class I evergreen and deciduous boreal forests. The regression equation for deciduous stands is not significant, and the equation for evergreen stands is $NPP_A = 104 + 36.7 (LAI)$, $r^2 = 0.20$, $P < 0.05$.

change," emphasizing the net exchange of carbon from a landscape or region is the sum of the product of net ecosystem exchange and area of each land cover or stage of succession.

Developing a boreal NPP database for ecosystem model validation

One objective of our paper was to determine if algorithms could be developed to estimate NPP from easily measured and readily available components of NPP, such as NPP_A or wood NPP. The extremely low number of complete NPP budgets and their nonrandom representation is a major limitation to testing NPP, and hence NEE, prediction of ecosystem process models for boreal forests—a region of great interest for global change. NPP_A was positively correlated to mean annual increment of aboveground biomass and mean annual total biomass increment for Class I stands (Fig. 2a). The positive relationship between NPP_A and mean annual biomass increment is especially useful because mean annual increment of aboveground biomass is a common variable that can be easily calculated from standard forest mensuration data available worldwide. The empirical model requires further evaluation, especially since there are potential autocorrelation problems.

Using a subset of 52 Class III stands that had NPP_A , NPP_B , and NPP, we observed a positive correlation ($r^2 = 0.96$, $n = 52$, $P < 0.01$) between the predicted NPP using the empirical relationship derived from Class I stands and the Class III NPP values compiled for these sites (data not shown). The high correlation suggests that the regression equation predictions are consistent with a large number of values reported in the literature, recognizing that Class III data represent a variety of methods and stand conditions that may not be equivalent to Class I stands.

Finally, the average NPP_A for the Class III stands (278

$g C \cdot m^{-2} \cdot yr^{-1}$) compared favorably to the average NPP for Class I stands ($257 g C \cdot m^{-2} \cdot yr^{-1}$), suggesting the 24 Class I stands are representative of boreal forests. However, the NPP_B for Class III stands is approximately half that of Class I stands (83 vs. $144 g C \cdot m^{-2} \cdot yr^{-1}$, respectively), confirming our suspicion that the NPP_B data for Class III stands are problematic. The empirical equation presented in Fig. 5 provides an approach to estimate NPP for Class III stands, increasing NPP data for boreal stands by 20-fold.

CONCLUSIONS AND RECOMMENDATIONS

Several intensive research programs such as Bonanza Creek LTER, BOREAS, and SWECON have greatly increased our understanding of the influence of environmental and ecological factors on NPP and carbon allocation patterns of boreal forests. Other individual studies have greatly increased NPP data for a variety of boreal forests that collectively make syntheses such as this study possible. Despite these valuable contributions, the number of complete NPP budgets and the geographic distribution of boreal stands are very limited. Major shortcomings of boreal forest NPP data include: (1) the lack of complete NPP budgets for boreal larch forests in Eurasia, (2) the lack of rigorous estimates of fine root NPP for most boreal forests, and (3) the lack of NPP data for understory and bryophyte vegetation.

We observed several empirical carbon allocation relationships that can be used to estimate NPP from annual woody biomass increment or NPP_A and establish upper and lower limits for carbon allocation belowground. The predictive power of these empirical relationships is greatly improved if the type of leaf habit is known. These empirical relationships provide one approach to increasing the NPP data for boreal forests that can be used to validate NPP estimates from process models.

A second problem related to extrapolating NPP from individual stands to larger areas is the influence of disturbance on NPP. Changes in species composition and

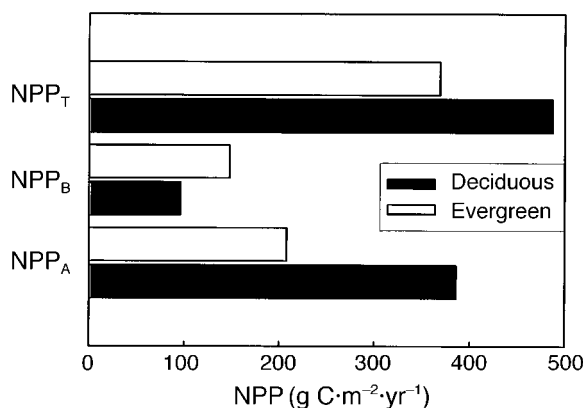


FIG. 5. Mean aboveground net primary production (NPP_A), belowground net primary production (NPP_B), and total net primary production (NPP_T) for Class I evergreen and deciduous boreal forests.

structure of boreal forests following disturbance influence NPP. Boreal forests are a mosaic of different-aged stands resulting from fire, logging, and insect damage, but our ability to model the effect of disturbance on the carbon budget of the boreal forests is limited by an incomplete understanding of the causes and magnitudes of the changes in NPP during succession.

Until the uncertainties of boreal forest NPP budgets are addressed, conclusions about whether the boreal forest is a carbon sink or source should be considered speculative.

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APPENDIX A

Summary of mean annual increment (MAI) and net primary production (NPP) for evergreen, deciduous, and combined (evergreen + deciduous) Class I, II, and III boreal forests.

Class and variable	N	Mean	SD	Minimum	Maximum
Class I, evergreens					
Stand age	18	99	59	20	250
MAI aboveground biomass	18	55	31	10	136
MAI total biomass	18	72	40	14	174
Wood NPP	18	145	100	35	323
Foliage NPP	18	69	26	30	125
Understory NPP	18	22	19	5	43
NPP _A	18	232	111	120	439
NPP _B	18	156	140	41	495
Total NPP	18	387	219	214	912
Class I, deciduous					
Stand age	7	58	21	30	90
MAI aboveground biomass	7	98	64	27	207
MAI total biomass	7	122	66	35	243
Wood NPP	7	301	141	176	480
Foliage NPP	7	103	31	65	155
Understory NPP	2	63	37	5	79
NPP _A	7	428	173	269	635
NPP _B	7	98	62	42	160
Total NPP	6	535	215	323	795
Class I, combined					
Stand age	24	87	54	20	250
MAI aboveground biomass	24	68	46	10	207
MAI total biomass	24	87	53	14	243
Wood NPP	24	181	124	35	480
Foliage NPP	24	80	31	30	155
Understory NPP	11	27	23	1	43
NPP _A	24	281	147	120	635
NPP _B	24	144	125	41	495
Total NPP	24	424	218	218	912
Class II, evergreen					
Stand age	37	69	47	14	260
MAI aboveground biomass	37	77	53	11	242
MAI total biomass	35	100	63	20	292
Wood NPP	32	155	137	3	542
Foliage NPP	33	93	41	4	148
Understory NPP	29	19	39	0	202
NPP _A	37	263	150	25	675
Class II, deciduous					
Stand age	8	79	77	29	250
MAI aboveground biomass	8	60	46	13	147
NPP _A	8	226	153	31	474
Class II, combined					
Stand age	45	71	52	14	260
MAI aboveground biomass	45	74	52	11	242
MAI total biomass	35	100	63	20	292
Wood NPP	37	160	130	3	542
Foliage NPP	38	98	41	4	171
Understory NPP	29	19	39	0	202
NPP _A	45	257	149	25	675
Class III					
Stand age	101	81	45	5	227
NPP _A	299	278	191	2	1030
NPP _B	52	83	73	1	315
Total NPP	52	362	236	46	1150

Notes: All measurements are in carbon grams per square meter per year, except for stand age, which is in years. NPP_A = aboveground net primary productivity; NPP_B = belowground net primary productivity. Data for Class III sites were compiled from Hall et al. (1992) and Esser et al. (1997).

APPENDIX B

Location, stand characteristics, and aboveground net primary production (NPP_A) for Class II boreal forest stands.

Country	Site	Species	Latitude, longitude
China	Daxing'anling	<i>Larix gmelinii</i>	49.00, 123.00
China	Daxing'anling	<i>Larix gmelinii</i>	50.00, 121.30
China	Daxing'anling	<i>Larix gmelinii</i>	50.00, 121.30
China	Daxing'anling	<i>Larix gmelinii</i>	52.20, 121.80
China	Daxing'anling	<i>Larix gmelinii</i>	52.20, 121.80
Russia	Koinas	<i>Picea abies</i>	64.67, 47.50
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia	Karelia	<i>Picea abies</i>	62.00, 34.00
Russia (Siberia)	Yakutsk	<i>Larix gmelinii</i>	60.85, 128.27
Russia (Siberia)	Yakutsk	<i>Larix gmelinii</i>	60.85, 128.27
Russia (Siberia)	Yakutsk	<i>Larix gmelinii</i>	60.85, 128.27
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Russia (Siberia)	Tomsk	<i>Pinus sylvestris</i>	58.00, 83.00
Sweden	Jädraås	<i>Pinus sylvestris</i>	60.82, 16.50
Finland	Oulu	<i>Picea excels</i>	66.37, 29.00
United States	Minnesota	<i>Picea mariana</i>	47.50, -93.50
United States	Minnesota	<i>Picea mariana</i>	47.50, -93.50
United States	Alaska	<i>Picea mariana</i>	64.00, -148.00
United States	Alaska	<i>Picea mariana</i>	64.00, -148.00
United States	Alaska	<i>Picea mariana</i>	64.00, -148.00

Note: The abbreviation "na" means "data not available."

† References: 1, Liu et al. 1994; 2, DeAngelis et al. 1981; 3, Kazimirov et al. 1977; 4, Schulze et al. 1995; 5, Gabeev 1990; 6, Agren et al. 1980; 7, Havas 1973; 8, Grigal et al. 1985.

APPENDIX B. Extended.

Mean annual precipitation (mm)	Mean annual temperature (°C)	Age (yr)	Aboveground biomass (g C/m ²)	NPP _A (g C·m ⁻² ·yr ⁻¹)	Source†
425	-4	29	4269	474	1
425	-4	54	3617	299	1
425	-4	29	2837	352	1
425	-4	55	2781	236	1
425	-4	34	2091	253	1
499	-1.2	125	6642	268	2
650	2.2	138	7703	127	3
650	2.2	37	1414	143	3
650	2.2	126	8304	154	3
650	2.2	22	1245	175	3
650	2.2	42	1725	184	3
650	2.2	109	7619	204	3
650	2.2	41	2088	216	3
650	2.2	45	1146	239	3
650	2.2	98	7386	240	3
650	2.2	37	2007	269	3
650	2.2	82	5505	285	3
650	2.2	45	2741	292	3
650	2.2	68	5051	308	3
650	2.2	54	3524	310	3
650	2.2	39	2228	314	3
650	2.2	43	2629	358	3
650	2.2	38	3031	389	3
213	-9.6	250	2011	31	4
213	-9.6	130	2272	58	4
213	-9.6	49	1070	107	4
501	na	122	16 338	229	5
501	na	25	2468	245	5
501	na	60	3493	260	5
501	na	70	9151	326	5
501	na	49	6552	339	5
501	na	80	7923	403	5
501	na	37	4070	417	5
501	na	31	7517	479	5
501	na	65	9893	523	5
501	na	67	11 086	601	5
501	na	47	10 395	675	5
731	3	14	156	56	6
500	0	260	5097	210	7
na	na	71	1796	143	8
na	na	106	5088	166	8
269	-3.4	51	1275	25	2
287	-3.4	55	1609	55	2
269	-3.4	130	9995	79	2